

0 Overview

Due Date: Monday, Sept 23 @ 8:00PM

Instructions

For this homework assignment you will define several functions. Each function accomplishes a specific task, but this week some functions may require you to call previously defined functions to get full credit. The functions are described below.

Before defining each function: come up with at least three test cases (other than the one provided). Before submitting to AutoLab, run your functions and check the results against your test cases.

All functions for this homework must be defined in a file named `main.py`. To submit and get feedback, submit this file to AutoLab under the Practice Problems – Week 4 submission. You may submit as many times as you like, but remember that AutoLab uses your LAST submission to determine your grade (not your highest).

These exercises are graded. You will benefit most from doing them entirely on your own, but of course the course staff is available to assist on Piazza and during office hours.

Learning Outcomes

You will demonstrate an ability to define and call functions in Python.

Supporting lecture topics

expressions simple expressions, compound expressions

statements assignment statement, return statement, import statement

functions function calls, function definitions

testing developing test cases

1 Problems

1. Define a function named `distance` that takes 4 numbers as parameters. The first two represent the x and y locations of a first point and the second two represent the x and y locations of a second point. The function needs to calculate and return the distance between those points. The calculation for this is $dist = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. The `math` module has a function named `sqrt` that takes a single argument and returns the square root of the argument.

Test Case: `distance(10,20,100,50)` should return 94.86832980505137.

2. Define a function named `getSpeed` that takes 2 parameters, representing a distance in miles and a time in minutes. The function should return a speed in miles per hour.

Test Case: `getSpeed(20,20)` should return 60.0.

3. Define a function named `gallonsPerHour` that takes 3 parameters, representing a distance in miles, a time in minutes, and the fuel efficiency of a car in miles per gallon. The function should return the number of gallons of fuel consumed per hour given the distance and time. (The speed of the car divided by the fuel efficiency will give you the gallons used per hour.)

Test Case: `gallonsPerHour(20,20,30)` should return 2.0.

Note: to get full credit for this function, it must call your `getSpeed` function to perform the calculation.

4. Define a function named `areaOfCircle` that takes 1 parameter representing the radius of a circle. The function should return the area of the circle. For reference, the area of a circle is $area = \pi * r^2$. The `math` module has a constant named `pi` that should be used for the value of π .

Test Case: `areaOfCircle(3)` should return 28.274333882308138.

5. Define a function named `volumeOfCylinder` that takes 2 parameters representing the radius and height of a cylinder in that order. The function should return the volume of the cylinder. For reference $volume = height * \pi * r^2$.

Test Case: `volumeOfCylinder(2,10)` should return 125.66370614359172.

Note: to get full credit for this function, it must call your `areaOfCircle` function to perform the calculation.

2 Submission

Save your `main.py` file and submit it to Week 4 in AutoLab.

Your final submission is due Monday, Sept 23 @ 8:00PM.